Fr(~si)
Fr(~pi)
Fr(~sr)
Fr(~pr)
Fr(~da)
Fr(~db)
Fr(~si2)
Fr(~pi2)
Fr(~sr2)
Fr(~pr2)
#vr : W3_Setup[W3Init( )]
!Subject(RE2, DO2, %1, ~si, ~pi, pk(~si), ppk(~pi))
!Subject('owner', DO1, %1, ~sr, ~pr, pk(~sr), ppk(~pr))
!Domain(DO2, %1, ~da, pk(~da))
!Domain(DO1, %1, ~db, pk(~db))
!Policy(DO2, DO1, %1, RE2, 'owner', '1')
!W3Ready(RE2, 'owner', DO2, DO1, %1)
!Subject(RE1, DO2, %1, ~si2, ~pi2, pk(~si2), ppk(~pi2))
!Subject('owner_2', DO1, %1, ~sr2, ~pr2, pk(~sr2), ppk(~pr2))
!Policy(DO2, DO1, %1, RE2, 'owner_2', '1')
!Policy(DO2, DO1, %1, RE1, 'owner', '1')
!Policy(DO2, DO1, %1, RE1, 'owner_2', '1')
!W3Ready(RE2, 'owner_2', DO2, DO1, %1)
!W3Ready(RE1, 'owner', DO2, DO1, %1)
!W3Ready(RE1, 'owner_2', DO2, DO1, %1)
Out(<pk(~si2), ppk(~pi2), %1, 'Subject Pk'>)
Out(<pk(~sr2), ppk(~pr2), %1, 'Subject Pk'>)
Out(< <pk(~si2), ppk(~pi2), %1>, SI4, pk(~da), SU1>)
Out(< <pk(~sr2), ppk(~pr2), %1>, sign(<pk(~sr2), ppk(~pr2), %1>, ~db), pk(~db), SU1>)
W3Budget( )
W3Budget( )
Out(<pk(~si), ppk(~pi), %1, 'Subject Pk'>)
Out(<pk(~sr), ppk(~pr), %1, 'Subject Pk'>)
Out(< <pk(~si), ppk(~pi), %1>, SI5, pk(~da), SU1>)
Out(< <pk(~sr), ppk(~pr), %1>, sign(<pk(~sr), ppk(~pr), %1>, ~db), pk(~db), SU1>)
Authentication_and_Data_Sharing_Session_2
!W3Ready(RE2, 'owner', DO2, DO1, %1)
W3Budget( )[+]
Fr(~rid)
!Subject(RE2, DO2, %1, ~si, ~pi, pk(~si), ppk(~pi))
!Subject('owner', DO1, %1, ~sr, ~pr, pk(~sr), ppk(~pr))
!Domain(DO2, %1, ~da, pk(~da))
!Domain(DO1, %1, ~db, pk(~db))
In(< <pk(~si), ppk(~pi), %1>, SI5, pk(~da), SU1>)
#b1 : Subject_Access_Request[Eq(SU1, SU1), Eq(pk(~da), pk(~da)), Eq(true, true), Eq(pk(~si), pk(~si)), Eq(ppk(~pi), ppk(~pi)), Eq(%1, %1), W3Begin(~rid, RE2, 'owner', %1), AccessRequest(RE2, 'owner', %1, DO2, DO1, ~si, ~sr, ~da)]
Out(< <<<pk(~si), ppk(~pi), %1>, SI5, pk(~da)>, pk(~sr), %1>, SI3, pk(~si), AC2>)
Access_send(RE2, 'owner', %1, pk(~si), pk(~sr), DO2, DO1, ~da, ~db, pk(~da), pk(~db), ppk(~pi), ~rid)
In(< < <<pk(~si), ppk(~pi), %1>, SI5, pk(~da)>, pk(~sr), %1>, SI3, pk(~si), AC2>)
Access_send(RE2, 'owner', %1, pk(~si), pk(~sr), DO2, DO1, ~da, ~db, pk(~da), pk(~db), ppk(~pi), ~rid)[+]
!Policy(DO2, DO1, %1, RE2, 'owner', '1')
!Subject('owner', DO1, %1, ~sr, ~pr, pk(~sr), ppk(~pr))
#vr.2 : Domain_Access_Verification[Eq(AC2, AC2), Eq(pk(~si), pk(~si)), Eq(true, true), Eq(pk(~sr), pk(~sr)), Eq(%1, %1), Eq('1', '1'), AccessGranted(RE2, 'owner', %1, pk(~si), pk(~sr), '1')]
Out(<H2, SI6, pk(~db), AC1>)
AccessCert_send(RE2, 'owner', %1, pk(~si), pk(~sr), ~sr, ppk(~pi), DO2, DO1, pk(~da), pk(~db), ~rid)
In(<H2, SI6, pk(~db), AC1>)
AccessCert_send(RE2, 'owner', %1, pk(~si), pk(~sr), ~sr, ppk(~pi), DO2, DO1, pk(~da), pk(~db), ~rid)[+]
!Subject(RE2, DO2, %1, ~si, ~pi, pk(~si), ppk(~pi))
!Subject('owner', DO1, %1, ~sr, ~pr, pk(~sr), ppk(~pr))
Fr(~x)
Fr(~k)
Fr(~v)
#vr.1 : Object_Data_Encryption[Eq(AC1, AC1), Eq(pk(~db), pk(~db)), Eq(true, true), Eq(H2, H2), W3Running(RE2, 'owner', %1, ~x, ~k), W3DataSent(RE2, 'owner', %1, ~x, PE2, SE2), DataSent('owner', RE2, %1, ~sr, ppk(~pi), pk(~si), ~x, ~k), Secret(~x, %1), SessionKey(~k, %1), Honest(RE2, %1), Honest('owner', %1)]
Out(<PE2, SE2, 'data'>)
X_send('owner', RE2, %1, ~sr, pk(~sr), ~pr, ~si, ~pi, <~x, sign(~x, ~sr), pk(~sr)>, ~x, ~k, ~rid)
In(<PE2, SE2, 'data'>)
X_send('owner', RE2, %1, ~sr, pk(~sr), ~pr, ~si, ~pi, <~x, sign(~x, ~sr), pk(~sr)>, ~x, ~k, ~rid)
#c1 : Subject_Data_Decryption[Eq('data', 'data'), Eq(~k, ~k), Eq(<~x, sign(~x, ~sr), pk(~sr)>, <~x, sign(~x, ~sr), pk(~sr)>), Eq(pk(~sr), pk(~sr)), Eq(true, true), Eq(~x, ~x), W3End(~rid, RE2, 'owner', %1), W3Commit(RE2, 'owner', %1, ~x, ~k), W3DataAccepted(RE2, 'owner', %1, ~x, PE2, SE2), DataReceive('owner', RE2, %1, PE2, SE2, ~pi, pk(~sr), ~k, ~x)]
Authentication_and_Data_Sharing_Session_1
!W3Ready(RE1, 'owner', DO2, DO1, %1)
W3Budget( )[+]
Fr(~rid.1)
!Subject(RE1, DO2, %1, ~si2, ~pi2, pk(~si2), ppk(~pi2))
!Subject('owner', DO1, %1, ~sr, ~pr, pk(~sr), ppk(~pr))
!Domain(DO2, %1, ~da, pk(~da))
!Domain(DO1, %1, ~db, pk(~db))
In(< <pk(~si2), ppk(~pi2), %1>, SI4, pk(~da), SU1>)
#b2 : Subject_Access_Request[Eq(SU1, SU1), Eq(pk(~da), pk(~da)), Eq(true, true), Eq(pk(~si2), pk(~si2)), Eq(ppk(~pi2), ppk(~pi2)), Eq(%1, %1), W3Begin(~rid.1, RE1, 'owner', %1), AccessRequest(RE1, 'owner', %1, DO2, DO1, ~si2, ~sr, ~da)]
Out(< < <<pk(~si2), ppk(~pi2), %1>, SI4, pk(~da) >, pk(~sr), %1>, SI1, pk(~si2), AC2>)
Access_send(RE1, 'owner', %1, pk(~si2), pk(~sr), DO2, DO1, ~da, ~db, pk(~da), pk(~db), ppk(~pi2), ~rid.1)
In(< < <<pk(~si2), ppk(~pi2), %1>, SI4, pk(~da)>, pk(~sr), %1>, SI1, pk(~si2), AC2>)
Access_send(RE1, 'owner', %1, pk(~si2), pk(~sr), DO2, DO1, ~da, ~db, pk(~da), pk(~db), ppk(~pi2), ~rid.1)[+]
!Policy(DO2, DO1, %1, RE1, 'owner', '1')
!Subject('owner', DO1, %1, ~sr, ~pr, pk(~sr), ppk(~pr))
#vr.4 : Domain_Access_Verification[Eq(AC2, AC2), Eq(pk(~si2), pk(~si2)), Eq(true, true), Eq(pk(~sr), pk(~sr)), Eq(%1, %1), Eq('1', '1'), AccessGranted(RE1, 'owner', %1, pk(~si2), pk(~sr), '1')]
Out(<H1, SI7, pk(~db), AC1>)
AccessCert_send(RE1, 'owner', %1, pk(~si2), pk(~sr), ~sr, ppk(~pi2), DO2, DO1, pk(~da), pk(~db), ~rid.1)
In(<H1, SI7, pk(~db), AC1>)
AccessCert_send(RE1, 'owner', %1, pk(~si2), pk(~sr), ~sr, ppk(~pi2), DO2, DO1, pk(~da), pk(~db), ~rid.1)[+]
!Subject(RE1, DO2, %1, ~si2, ~pi2, pk(~si2), ppk(~pi2))
!Subject('owner', DO1, %1, ~sr, ~pr, pk(~sr), ppk(~pr))
Fr(~x.1)
Fr(~k.1)
Fr(~v.1)
#vr.3 : Object_Data_Encryption[Eq(AC1, AC1), Eq(pk(~db), pk(~db)), Eq(true, true), Eq(H1, H1), W3Running(RE1, 'owner', %1, ~x.1, ~k.1), W3DataSent(RE1, 'owner', %1, ~x.1, PE1, SE1), DataSent('owner', RE1, %1, ~sr, ppk(~pi2), pk(~si2), ~x.1, ~k.1), Secret(~x.1, %1), SessionKey(~k.1, %1), Honest(RE1, %1), Honest('owner', %1)]
Out(<PE1, SE1, 'data'>)
X_send('owner', RE1, %1, ~sr, pk(~sr), ~pr, ~si2, ~pi2, <~x.1, sign(~x.1, ~sr), pk(~sr)>, ~x.1, ~k.1, ~rid.1)
In(<PE1, SE1, 'data'>)
X_send('owner', RE1, %1, ~sr, pk(~sr), ~pr, ~si2, ~pi2, <~x.1, sign(~x.1, ~sr), pk(~sr)>, ~x.1, ~k.1, ~rid.1)
#c2 : Subject_Data_Decryption[Eq('data', 'data'), Eq(~k.1, ~k.1), Eq(<~x.1, sign(~x.1, ~sr), pk(~sr)>, <~x.1, sign(~x.1, ~sr), pk(~sr)>), Eq(pk(~sr), pk(~sr)), Eq(true, true), Eq(~x.1, ~x.1), W3End(~rid.1, RE1, 'owner', %1), W3Commit(RE1, 'owner', %1, ~x.1, ~k.1), W3DataAccepted(RE1, 'owner', %1, ~x.1, PE1, SE1), DataReceive('owner', RE1, %1, PE1, SE1, ~pi2, pk(~sr), ~k.1, ~x.1)]
#vl.6 : irecv
#vk.15 : coerce[!KU(H2)]
#vr.15 : d_0_snd
#vl.2 : irecv
#vk.2 : coerce[!KU(PE2)]
#vr.21 : d_0_snd
#vk.6 : coerce[!KU(SI6)]
#vr.32 : d_0_snd
#vk.9 : coerce[!KU(SE2)]
#vk.16 : coerce[!KU(pk(~db))]
#vf.4 : isend
#vf.16 : isend
#vl.7 : irecv
#vr.17 : d_0_snd
#vk.18 : coerce[!KU(H1)]
#vk.7 : coerce[!KU(SI7)]
#vf.21 : isend
#vl.5 : irecv
#vl.4 : irecv
#vr.13 : d_0_snd
#vr.11 : d_0_snd
#vr.19 : d_0_snd
#vl.8 : irecv
#vl.9 : irecv
#vk.4 : coerce[!KU(SI5)]
#vk.11 : coerce[!KU(pk(~si))]
#vk.12 : coerce[!KU(ppk(~pi))]
#vk.8 : coerce[!KU(pk(~da))]
#vk.13 : coerce[!KU(pk(~si2))]
#vk.14 : coerce[!KU(ppk(~pi2))]
#vk.5 : coerce[!KU(SI4)]
#vf.1 : isend
#vf.3 : isend
#vf.20 : isend
#vl : irecv
#vr.34 : d_0_fst
#vl.1 : irecv
#vl.3 : irecv
#vr.5 : d_0_snd
#vr.35 : d_0_snd
#vr.7 : d_0_snd
#vk.3 : coerce[!KU(PE1)]
#vr.23 : d_0_snd
#vk : coerce[!KU(SI3)]
#vk.17 : coerce[!KU(pk(~sr))]
#vk.1 : coerce[!KU(SI1)]
#vk.10 : coerce[!KU(SE1)]
#vf.25 : isend
#vf.5 : isend
AC1 = 'AccessCert_ab'
AC2 = 'AccessR_a'
DO1 = 'domain_B'
DO2 = 'domain_A'
H1 = h(<pk(~si2), pk(~sr), %1>)
H2 = h(<pk(~si), pk(~sr), %1>)
PE1 = PEnc(ppk(~pi2), ~k.1, ~v.1)
PE2 = PEnc(ppk(~pi), ~k, ~v)
RE1 = 'requester_2'
RE2 = 'requester'
SE1 = senc(<~x.1, sign(~x.1, ~sr), pk(~sr)>, ~k.1)
SE2 = senc(<~x, sign(~x, ~sr), pk(~sr)>, ~k)
SI4 = sign(<pk(~si2), ppk(~pi2), %1>, ~da)
SI1 = sign(<<<pk(~si2), ppk(~pi2), %1>, SI4, pk(~da)>, pk(~sr), %1>, ~si2)
SI5 = sign(<pk(~si), ppk(~pi), %1>, ~da)
SI3 = sign(<<<pk(~si), ppk(~pi), %1>, SI5, pk(~da)>, pk(~sr), %1>, ~si)
SI6 = sign(H2, ~db)
SI7 = sign(H1, ~db)
SU1 = 'SubjectCert'